

Credit Risk Decision Engine

End-to-End Machine Learning Project Report

Final Frozen Version

An end-to-end credit-risk system that converts customer information into calibrated probability of default, operational lending decisions, risk segmentation, portfolio expected-loss analysis, explainability and a validated deployment pipeline.

Component	Status
Data preparation	Complete
Feature engineering	Complete
Model development	Complete
Probability calibration	Complete
Decision engine	Complete
Risk segmentation	Complete
Expected-loss analysis	Complete
SHAP explainability	Complete
Deployment validation	Complete
Stress testing	Complete
Governance & audit	Complete

Frozen-model statement: The model, preprocessing, calibration, decision thresholds and official evaluation results documented in this report are unchanged by the documentation update.

Executive Summary

The project implements an end-to-end credit risk decision system using a LightGBM classifier followed by probability calibration. The calibrated probability of default (PD) is converted into APPROVE, REVIEW and REJECT lending actions and four operational risk bands. The system additionally estimates expected financial loss using $PD \times Exposure \times LGD$ and includes a packaged deployment pipeline with stress and governance validation.

The project uses 172,205 training customers and 43,052 held-out test customers. The feature pipeline starts with 99 original input columns and engineers seven additional financial/behavioral features, producing 106 raw/engineered inputs. The verified preprocessing pipeline transforms these inputs into 226 model features.

The final LightGBM model achieved 0.7395 ROC-AUC and 0.2264 PR-AUC on the held-out test set. Five-fold OOF validation produced 0.7340 ± 0.0018 ROC-AUC. SHAP TreeExplainer was applied to 5,000 held-out observations across all 226 model features for global feature importance and risk-direction analysis.

1. Business Problem

A credit-risk system should not stop at a binary classifier. It should estimate default probability, make that probability usable for risk decisions, translate it into operational lending actions, segment customers by risk and quantify potential portfolio loss.

Objectives: predict probability of default; calibrate probabilities; map calibrated PD to APPROVE/REVIEW/REJECT; create four risk bands; estimate expected loss and risk concentration; provide global SHAP explainability; validate the complete inference pipeline and edge-case behavior.

2. Dataset and Feature Pipeline

The project uses the Home Credit Default Risk dataset. The raw dataset is not included in the repository.

Item	Final value
Training customers	172,205
Held-out test customers	43,052
Original input columns	99
Engineered input features	7
Raw/engineered inputs	106
Processed model features	226
Observed test default rate	8.11%

Feature engineering: CREDIT_INCOME_RATIO, ANNUITY_INCOME_RATIO, CREDIT_GOODS_RATIO, INCOME_PER_FAMILY_MEMBER, CREDIT_PER_FAMILY_MEMBER, EMPLOYMENT_INCOME_RATIO and EXT_SOURCE_MEAN.

The project contains 92 numerical and 14 categorical inputs before preprocessing. The reconstructed preprocessing pipeline was numerically verified against the original processed matrices, producing identical training and test transformations and preserving the 226-feature model interface.

3. Exploratory Analysis and Feature Engineering

The analysis examined missingness, feature distributions and target-level differences for engineered financial relationships. The engineered features express affordability, credit burden, family-adjusted financial capacity and external-score information in a form suitable for the model.

4. Train/Test Split and Preprocessing

The modeling data contains 172,205 training customers and a held-out test set of 43,052 customers. Ninety-two numerical and fourteen categorical columns are passed through the preprocessing pipeline, producing 226 model

features.

The original transformation logic was reconstructed and verified against the stored processed matrices with zero maximum and mean numerical difference, after which the verified preprocessor was packaged.

5. Model Development

A Logistic Regression model was used as a baseline before the final LightGBM model.

The final predictive model is LightGBM. It generates raw default probabilities from the 226 processed features. A separate one-feature calibration model converts the raw model probability into calibrated PD for business use.

Model	ROC-AUC	PR-AUC
Logistic Regression baseline	0.7260	0.2035
LightGBM final model	0.7395	0.2264

6. Model Validation and Generalization

The five OOF folds show relatively small variation, with ROC-AUC values from 0.7318 to 0.7364. The train-test gap of 0.1176 is retained as the principal model limitation because it indicates remaining overfitting.

Metric	Result
Test ROC-AUC	0.7395
Test PR-AUC	0.2264
Calibrated Brier Score	0.0691
Train ROC-AUC	0.8571
Train-Test Gap	0.1176
OOF ROC-AUC	0.7340
OOF Std	0.0018
OOF minimum / maximum	0.7318 / 0.7364
OOF fold range	0.0046

7. Probability Calibration

Raw LightGBM probabilities are passed to a one-feature calibration model. The calibrated PD is used for downstream lending decisions rather than the raw score. Deployment validation explicitly checks that the calibration model receives one raw-probability feature.

Calibration measure	Result
Calibrated Brier Score	0.0691
Maximum absolute calibration gap	0.0095
Mean absolute calibration gap	0.0048

8. Explainability — SHAP

Global SHAP explainability is completed in the final notebook. SHAP TreeExplainer was applied to a representative sample of 5,000 held-out observations across all 226 model features. Mean absolute SHAP values were used to rank global feature importance, and mean SHAP values were used to identify overall direction of contribution.

The SHAP analysis is post-hoc explainability; it did not change the trained model, thresholds or official performance results. Transformed features such as Column_91 and Column_24 are not assigned business meanings without a verified feature mapping. SHAP contribution is model explanation, not causal evidence.

Rank	Feature	Mean SHAP	Mean SHAP	Direction
1	Column_91	0.294596	-0.003784	Decreases risk
2	Column_24	0.187688	0.007216	Increases risk
3	Column_87	0.143132	0.000381	Increases risk
4	Column_25	0.132394	0.004953	Increases risk
5	Column_4	0.125018	0.008557	Increases risk
6	Column_90	0.110075	-0.010760	Decreases risk
7	Column_5	0.098401	-0.002514	Decreases risk
8	Column_23	0.097755	0.009824	Increases risk
9	Column_3	0.089441	0.040596	Increases risk
10	Column_7	0.088040	-0.004938	Decreases risk

9. Decision Engine

Calibrated PD is converted into three operational lending actions using frozen business rules. The held-out test population contains 30,873 APPROVE, 7,412 REVIEW and 4,767 REJECT decisions. Decision-level observed default rates increase from APPROVE to REVIEW to REJECT, and all customers receive exactly one decision.

Decision	Rule	Customer share
APPROVE	$PD < 0.10$	71.71%
REVIEW	$0.10 \leq PD < 0.18$	17.22%
REJECT	$PD \geq 0.18$	11.07%

10. Risk Segmentation

Observed default rates increase monotonically across the four risk bands, supporting the intended ordering of the risk segmentation.

Risk band	Customers	Average PD	Observed default rate	Share
Low Risk	18,987	0.0313	0.0289	44.10%
Medium Risk	11,886	0.0711	0.0727	27.61%
High Risk	7,412	0.1340	0.1218	17.22%
Very High Risk	4,767	0.2441	0.2465	11.07%

11. Expected Loss and Business Impact

Expected loss is calculated as $PD \times Exposure \times LGD$. The exposure variable is AMT_CREDIT and the analysis uses an LGD assumption of 0.45.

The central business finding is that 28.29% of customers account for approximately 55.96% of modeled expected loss. High-risk customers therefore contribute a disproportionately large share of portfolio risk.

Portfolio measure	Final value
Customers	43,052
Total exposure	25,974,998,023.50
Average PD	0.0835
Total expected loss	905,828,407.72
Expected loss rate	3.4873%
High + Very High customer share	28.29%

Portfolio measure	Final value
High + Very High expected-loss share	55.96%
Very High expected-loss share	28.98%

12. Deployment Pipeline

The project packages the preprocessing pipeline, LightGBM model, calibration model, feature metadata, decision configuration and model metadata. Reload tests reproduced the original raw probabilities and calibrated PDs without numerical difference.

Inference flow: 106 raw/engineered inputs → preprocessing → 226 model features → LightGBM raw PD → probability calibration → calibrated PD → decision and risk bands.

A single-customer prediction interface and multi-customer batch interface were validated.

13. Stress and Edge-Case Testing

The deployment pipeline passed single-customer, 100-customer batch, missing-value, extreme numerical value, unseen categorical value, deterministic repeated prediction, threshold-logic and invalid-feature-count tests. The full deployment dataset also passed PD-range, NaN/infinite-value, decision-logic, risk-band, threshold-boundary and customer-count reconciliation checks.

Test	Status
Single customer	PASS
100-customer batch	PASS
Missing values	PASS
Extreme numerical values	PASS
Unseen categorical value	PASS
Deterministic repeated prediction	PASS
Decision threshold logic	PASS
Invalid feature-count protection	PASS

14. Governance and Audit

The final governance audit covers model performance, OOF/generalization, decision-engine configuration, business impact, deployment artifacts and stress testing. Governance completeness was recorded as 100% with an EXCELLENT grade.

Governance component	Status
Performance audit	Complete
OOF/generalization audit	Complete
Decision-engine audit	Complete
Business-impact audit	Complete
Deployment audit	Complete
Stress testing	Complete
Governance completeness	100%

15. Final Scorecard

Metric	Final result
Test ROC-AUC	0.7395
Test PR-AUC	0.2264
Brier Score	0.0691
OOF ROC-AUC $\pm$ Std	0.7340 $\pm$ 0.0018
Train ROC-AUC	0.8571
Train-Test Gap	0.1176
Customers evaluated	43,052
Expected loss	905,828,407.72
Expected loss rate	3.4873%
High/Very High customer share	28.29%
High/Very High expected-loss share	55.96%
Governance completeness	100%

16. Limitations

The train-test ROC-AUC gap of 0.1176 indicates remaining overfitting.

Test ROC-AUC of 0.7395 leaves room for stronger discrimination through model comparison and tuning.

The decision thresholds are validated business rules; they are not claimed to be globally cost-optimal.

Global SHAP analysis is completed, but a full customer-level/local explanation interface is not implemented.

The deployment pipeline is validated offline; live production monitoring for drift, calibration degradation and performance is not implemented.

17. Future Improvements

Stronger regularization and validation to reduce the generalization gap.

Alternative model and hyperparameter comparison.

Additional domain-specific feature engineering.

Cost-sensitive threshold optimization using explicit financial costs.

Customer-level/local SHAP explanations.

Production drift, calibration and performance monitoring.

Final Project Status

Project 1 is documented as complete and frozen. The official model, preprocessing, calibration, decision thresholds and evaluation results remain unchanged by this report.